



Better days

JURIS NOVA

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À PROPOS DE NOUS



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Marketing agent and technical support



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INTRODUCTION

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segmentée et sécurisée

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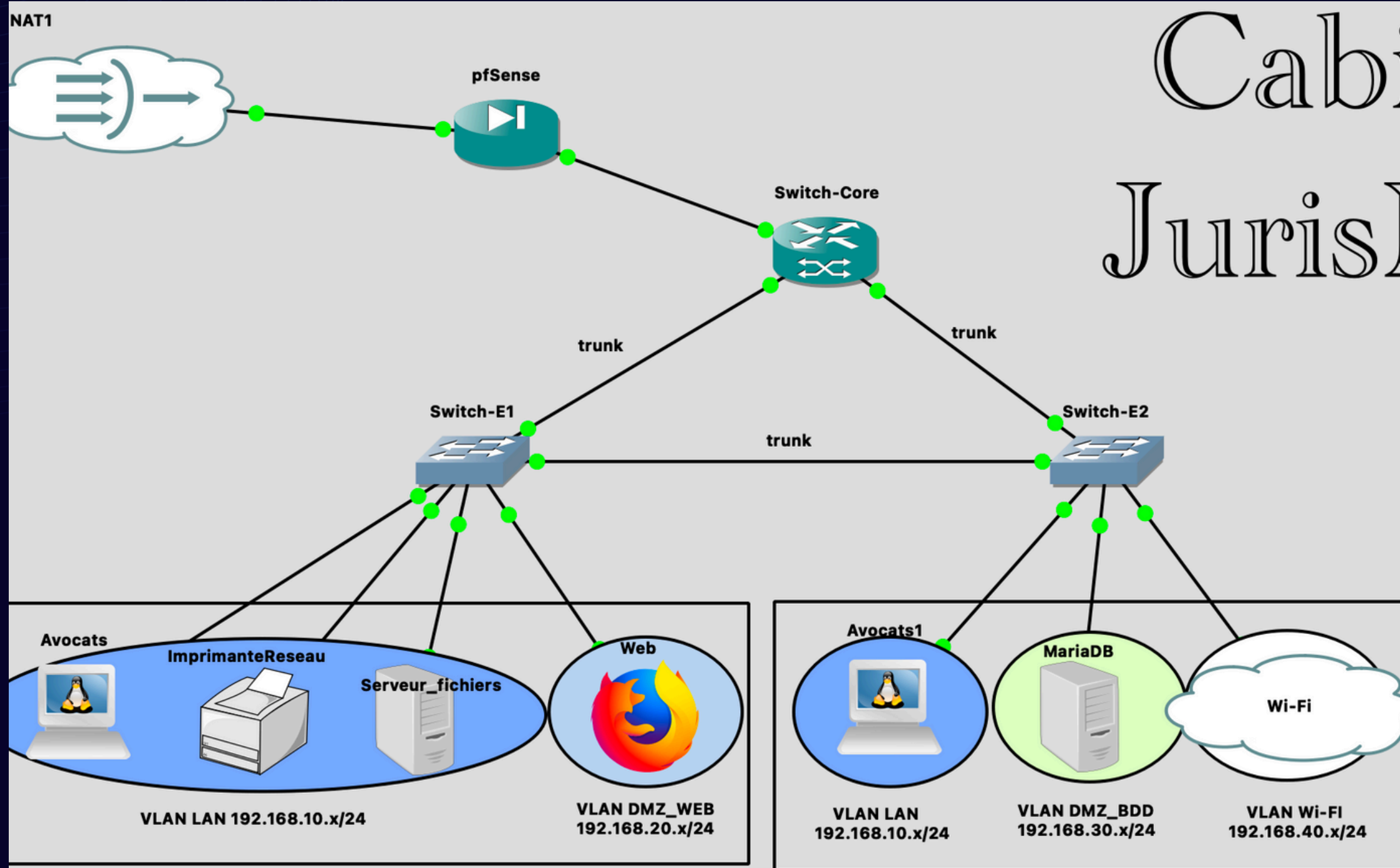
Fonctionnement en conditions
réelles



ARCHITECTURE RÉSEAU SEGMENTÉE ET SÉCURISÉE

- 2 étages reliés via double gaine technique
- Répartition en VLAN :
- DMZ Web (site vitrine, GLPI)
- DMZ BDD (MariaDB, accès restreint)
- LAN (postes, fichiers, imprimante)
- Wi-Fi invité (isolé, accès Internet seul)
- STP activé pour éviter les boucles

Cabinet JurisNova



```
• • • andonpendichev — Avocats — telnet 172.16.75.129 5034 — 80x24
Avocats : 192.168.10.1 255.255.255.0 gateway 192.168.10.254

[Avocats> ping 192.168.10.254

192.168.10.254 icmp_seq=1 timeout
192.168.10.254 icmp_seq=2 timeout
192.168.10.254 icmp_seq=3 timeout
84 bytes from 192.168.10.254 icmp_seq=4 ttl=255 time=354.135 ms
192.168.10.254 icmp_seq=5 timeout

[Avocats> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=0.876 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=1.096 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=1.521 ms
^C
[Avocats> ping 192.168.10.4

84 bytes from 192.168.10.4 icmp_seq=1 ttl=64 time=0.960 ms
84 bytes from 192.168.10.4 icmp_seq=2 ttl=64 time=685.582 ms
192.168.10.4 icmp_seq=3 timeout
84 bytes from 192.168.10.4 icmp_seq=4 ttl=64 time=0.593 ms
84 bytes from 192.168.10.4 icmp_seq=5 ttl=64 time=0.540 ms
```

```
[MariaDB> ping 192.168.30.1

192.168.30.1 icmp_seq=1 ttl=64 time=0.001 ms
192.168.30.1 icmp_seq=2 ttl=64 time=0.001 ms
192.168.30.1 icmp_seq=3 ttl=64 time=0.001 ms
192.168.30.1 icmp_seq=4 ttl=64 time=0.001 ms
192.168.30.1 icmp_seq=5 ttl=64 time=0.001 ms

[MariaDB> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=63 time=604.552 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=63 time=660.346 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=63 time=654.151 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=63 time=756.459 ms
```

```
R1#pin
*Jun 17 11:04:24.667: %SYS-5-CONFIG_I: Configured from console by console
[R1#ping 192.168.30.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 252/306/380 ms
[R1#ping 192.168.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 320/372/420 ms
R1#
```

```
• • • andonpendichev — Routeur — telnet 172.16.75.129 5032 — 80x24
1, changed state to up
R1(config-if)#Interface Ethernet0/1 assigned DHCP address 192.168.122.60, mask 2
55.255.255.0

[R1(config-if)#exit
[R1(config)#do ping 8.8.8.8

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/29/48 ms
[R1(config)#do ping 192.168.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 296/368/576 ms
```

andonpendichev — Switch-E1 — telnet 172.16.75.130 5021 — 80x24

```
[VLAN0020
[ Spanning tree enabled protocol ieee
[ Root ID      Priority      32788
[              Address      aabb.cc00.0100
[              This bridge is the root
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[
[ Bridge ID    Priority      32788 (priority 32768 sys-id-ext 20)
[              Address      aabb.cc00.0100
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[              Aging Time    15 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Desg	LRN	100	128.1	Shr
Et0/1	Desg	LRN	100	128.2	Shr
Et0/3	Desg	LRN	100	128.4	Shr

Switch-E1#

andonpendichev — Switch-E1 — telnet 172.16.75.130 5021 — 80x24

```
[VLAN0020
[ Spanning tree enabled protocol ieee
[ Root ID      Priority      32788
[              Address      aabb.cc00.0100
[              This bridge is the root
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[
[ Bridge ID    Priority      32788 (priority 32768 sys-id-ext 20)
[              Address      aabb.cc00.0100
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[              Aging Time    15 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Desg	LRN	100	128.1	Shr
Et0/1	Desg	LRN	100	128.2	Shr
Et0/3	Desg	LRN	100	128.4	Shr

Switch-E1#

Switch-E1#

Switch-E1#

andonpendichev — Switch-Core — telnet 172.16.75.130 5024 — 80x24

```
[Et0/2          Desg FWD 100      128.3  Shr
```

```
[VLAN0020
[ Spanning tree enabled protocol ieee
[ Root ID      Priority      32788
[              Address      aabb.cc00.0100
[              Cost          100
[              Port          2 (Ethernet0/1)
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[
[ Bridge ID    Priority      32788 (priority 32768 sys-id-ext 20)
[              Address      aabb.cc00.0200
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[              Aging Time    300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Desg	FWD	100	128.1	Shr
Et0/1	Root	FWD	100	128.2	Shr
Et0/2	Desg	FWD	100	128.3	Shr

```
[VLAN0030
[ Spanning tree enabled protocol ieee
[ Root ID      Priority      32798
[              Address      aabb.cc00.0200
[              This bridge is the root
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[
[ Bridge ID    Priority      32798 (priority 32768 sys-id-ext 30)
[              Address      aabb.cc00.0200
[              Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
[              Aging Time    300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Desg	FWD	100	128.1	Shr
Et0/1	Desg	FWD	100	128.2	Shr
Et0/2	Desg	FWD	100	128.3	Shr

SYNTHÈSE DES ÉQUIPEMENTS

BETTER DAYS
22, Avenue Voltaire
Villetaneuse

Destinataire
JurisNova
31, rue de la Forêt
75008 Paris



DEVIS

Référence du devis : 123
Date du devis : 19/06/2025
Date de validité du devis : 03/07/2025

Informations additionnelles :

Service après-vente - Garantie : 1 an.
Date de début de la prestation : 14/06/2025

Description	Quantité	Unité	Prix unitaire	% de TVA	Total TVA	Total TTC
PC : Dell Optiplex 7010	12	Unité	700,00 €	20 %	1 680,00 €	10 080,00 €
Serveur : Fichier	1	Unité	1 100,00 €	20 %	220,00 €	1 320,00 €
Serveur Web : ProLiant ML30 G10	1	Unité	1 000,00 €	10 %	100,00 €	1 100,00 €
Serveur BDD : HP ProLiant DL380 Gen10	1	Unité	1 500,00 €	10 %	150,00 €	1 650,00 €
Imprimantes Réseau : Brother HL-L6400DW	1	Unité	350,00 €	10 %	35,00 €	385,00 €
Pare-feu : pfsense Netgate SG-2100	1	Unité	500,00 €	10 %	50,00 €	550,00 €
Switch N2 : Cisco catalyst C1000-24T-4G-L	1	Unité	450,00 €	10 %	45,00 €	495,00 €
Switch Core : Cisco catalyst 3560CX-12PC-S	1	Unité	1 000,00 €	10 %	100,00 €	1 100,00 €
Point d'accès Wifi Pro : Ubiquiti UniFi 6 Lite	1	Unité	150,00 €	10 %	15,00 €	165,00 €
Fibre d'optique & RJ45 : Kit cablages + modules SFP		lot				500,00 €
Onduleur,baie de brassage, support		lot				700,00 €

Total HT 15 650,00 €
Total TVA 2 395,00 €

Total TTC 18 045,00 €

MESURES DE SÉCURITÉ ET HAUTE DISPONIBILITÉ

- VLAN isolé : Wi-Fi
- Aucun accès WAN
vers Web ni vers
MariaDB

```

Routeur#sh access-list
Standard IP access list NAT_ACL
    10 permit 192.168.10.0, wildcard bits 0.0.0.255
    20 permit 192.168.20.0, wildcard bits 0.0.0.255
    30 permit 192.168.30.0, wildcard bits 0.0.0.255
    40 permit 192.168.40.0, wildcard bits 0.0.0.255 (3 matches)
Extended IP access list DMZ_PROTECTION
[
    10 permit ip 192.168.20.0 0.0.0.255 host 192.168.20.254
    20 permit ip 192.168.30.0 0.0.0.255 host 192.168.30.254
    30 deny ip any 192.168.20.0 0.0.0.255
    40 deny ip any 192.168.30.0 0.0.0.255
    50 permit ip any any
Extended IP access list WIFI_ISOLATION
    10 permit ip 192.168.40.0 0.0.0.255 host 192.168.40.254
[
    20 permit udp 192.168.40.0 0.0.0.255 any eq domain
    30 permit icmp 192.168.40.0 0.0.0.255 host 192.168.40.254
    40 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
    50 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255 (9 matches)
    60 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255 (15 matches)
    70 permit ip 192.168.40.0 0.0.0.255 any (3 matches)
Routeur#

```

```

Wi-Fi> ping 192.168.20.1

*192.168.40.254 icmp_seq=1 ttl=255 time=9.084 ms (ICMP type:3, code:13, Communication administratively prohibited)
[*192.168.40.254 icmp_seq=2 ttl=255 time=11.200 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.40.254 icmp_seq=3 ttl=255 time=10.257 ms (ICMP type:3, code:13, Communication administratively prohibited)
^C
Wi-Fi> ping 192.168.30.1

*192.168.40.254 icmp_seq=1 ttl=255 time=11.281 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.40.254 icmp_seq=2 ttl=255 time=11.528 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.40.254 icmp_seq=3 ttl=255 time=1.935 ms (ICMP type:3, code:13, Communication administratively prohibited)
^X*192.168.40.254 icmp_seq=4 ttl=255 time=3.047 ms (ICMP type:3, code:13, Communication administratively prohibited)
[*192.168.40.254 icmp_seq=5 ttl=255 time=1.788 ms (ICMP type:3, code:13, Communication administratively prohibited)

Wi-Fi>
Wi-Fi> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=126 time=24.427 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=126 time=12.202 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=126 time=21.401 ms
^C
Wi-Fi>

```

```

Wi-Fi> ping 192.168.20.1

*192.168.40.254 icmp_seq=1 ttl=255 time=9.084 ms (ICMP type:3, code:13, Communication administratively prohibited)
[*192.168.40.254 icmp_seq=2 ttl=255 time=11.200 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.40.254 icmp_seq=3 ttl=255 time=10.257 ms (ICMP type:3, code:13, Communication administratively prohibited)
^C
Wi-Fi> ping 192.168.30.1

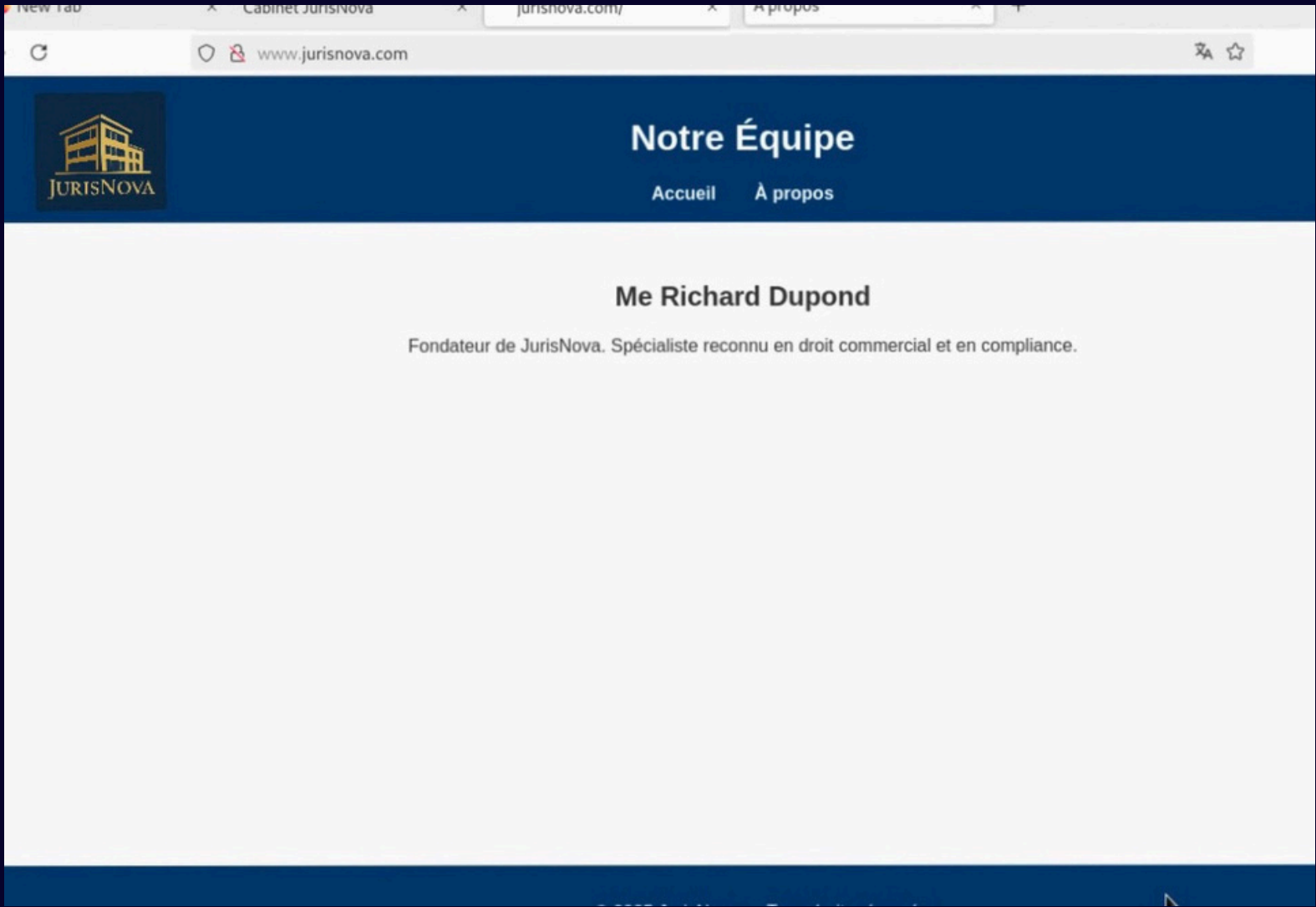
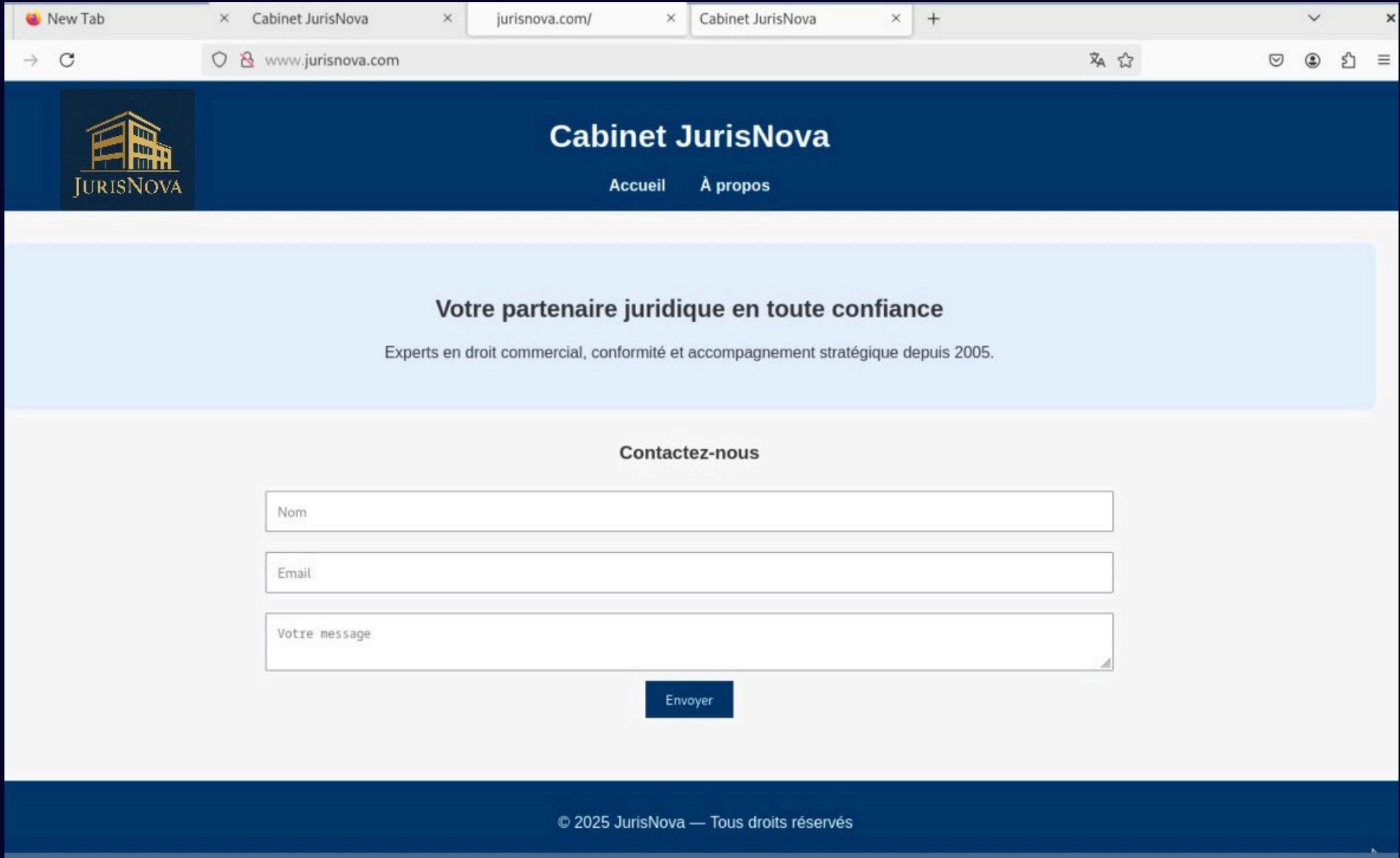
*192.168.40.254 icmp_seq=1 ttl=255 time=11.281 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.40.254 icmp_seq=2 ttl=255 time=11.528 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.40.254 icmp_seq=3 ttl=255 time=1.935 ms (ICMP type:3, code:13, Communication administratively prohibited)
^X*192.168.40.254 icmp_seq=4 ttl=255 time=3.047 ms (ICMP type:3, code:13, Communication administratively prohibited)
[*192.168.40.254 icmp_seq=5 ttl=255 time=1.788 ms (ICMP type:3, code:13, Communication administratively prohibited)

Wi-Fi>
Wi-Fi> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=126 time=24.427 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=126 time=12.202 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=126 time=21.401 ms
^C
Wi-Fi>

```

```
glpiuser@glpi: ~  
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:31:27:8e brd ff:ff:ff:ff:ff:ff  
    altname enp2s0  
    inet 172.16.75.132/24 brd 172.16.75.255 scope global dynamic noprefixroute ens160  
        valid_lft 1305sec preferred_lft 1305sec  
    inet6 fe80::20c:29ff:fe31:278e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
root@glpi:~# ping 192.168.30.2  
^[[A^[[A^[[B^[[B^[[B^[[A^[[A^C  
root@glpi:~# ping 192.168.30.2  
PING 192.168.30.2 (192.168.30.2) 56(84) bytes of data:  
64 bytes from 192.168.30.2: icmp_seq=1 ttl=64 time=0.130 ms  
64 bytes from 192.168.30.2: icmp_seq=2 ttl=64 time=0.112 ms  
64 bytes from 192.168.30.2: icmp_seq=3 ttl=64 time=0.103 ms  
64 bytes from 192.168.30.2: icmp_seq=4 ttl=64 time=0.119 ms  
64 bytes from 192.168.30.2: icmp_seq=5 ttl=64 time=0.102 ms  
64 bytes from 192.168.30.2: icmp_seq=6 ttl=64 time=0.105 ms  
^C  
--- 192.168.30.2 ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5123ms  
rtt min/avg/max/mdev = 0.102/0.111/0.130/0.010 ms  
root@glpi:~#
```





CONTACT US



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Ile de France

MERCI POUR VOTRE ATTENTION !
